More New Physics with Neutrinos

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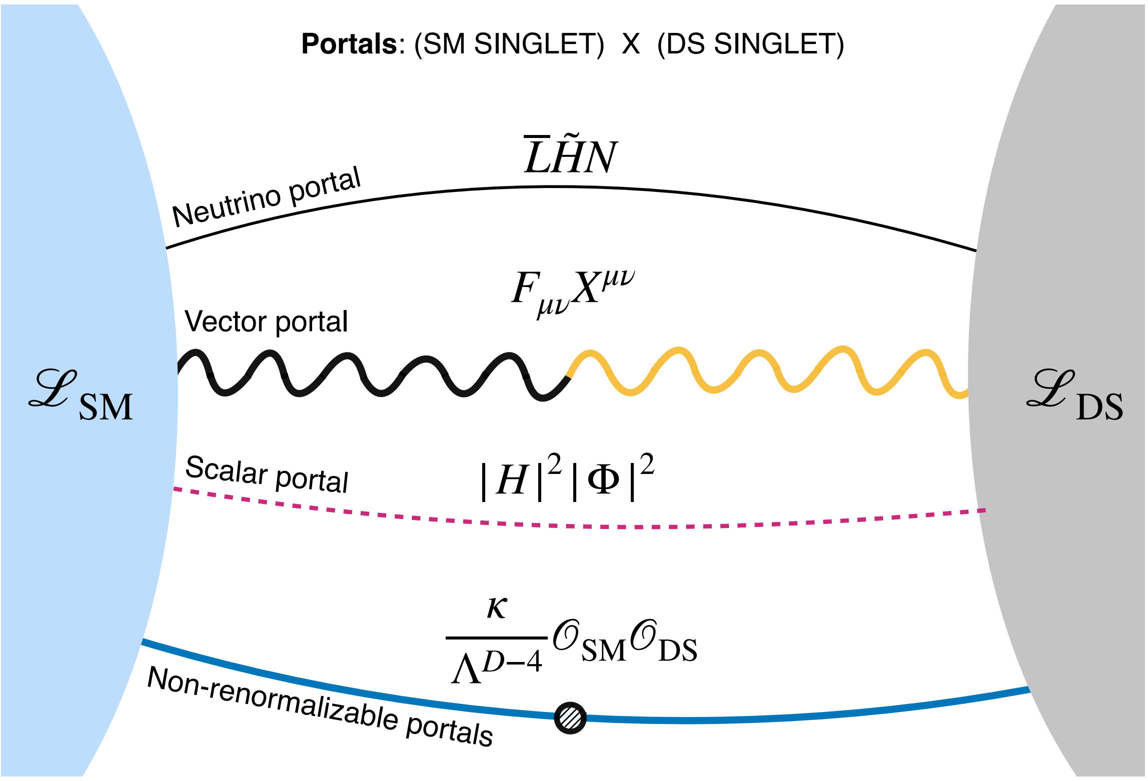

1 Why a Dark Sector?

The existence of dark matter is another strong motivation to consider physics beyond the Standard Model. There are many excellent reviews on the subject, so I will not belabor the point here.

The key for these lectures is that the dark matter problem is a strong motivation to consider the existence of a "dark sector": a sector of particles that is not charged under the Standard Model gauge group, but that may have its own interactions and particles. These dark sector-SM interactions should be very weak as otherwise we would have already seen them in experiments.

2 Renormalizable Portals

There is a nice framework to think about portals to dark sectors by the name of "portals". These are dimension-4 operators that connect the Standard Model to a dark sector. The idea is to write combinations of SM fields that are invariant under the SM gauge group and attach them to some new singlet field from the dark sector.

The simplest renormalizable portals are the following:

- The vector portal:
$$\mathcal{L} = \frac{\epsilon}{2} B_{\mu\nu} X^{\mu\nu}, \tag{2.1}$$
where $B_{\mu\nu}$ is the hypercharge field strength and $X^{\mu\nu}$ is the field strength of a new $U(1)_X$ gauge boson. This is often called the kinetic mixing portal, and X is the dark photon. You may also see the vector portal written in terms of the EM field strength tensor $F_{\mu\nu}X^{\mu\nu}$. These two forms lead to similar phenomenology, up to couplings of the dark photon to the neutral current that shows up in the $B_{\mu\nu}X^{\mu\nu}$ form, but not in the $F_{\mu\nu}X^{\mu\nu}$ form. The former is more general.

- The Higgs portal(s),
$$\mathcal{L} = \mu H^\dagger H S + \lambda H^\dagger H S^2 \tag{2.2}$$
where S is a new scalar that is a singlet under the Standard Model gauge group. The dimension-3 portal (super-renormalizable) comes with a dimensionful coupling μ, while the dimension-4 portal comes with a dimensionless coupling λ. Through mass mixing with the Higgs, the singlet scalar can couple to the rest of the SM through "weaker-than-Yukawa" couplings. Sometimes S is called a "dark Higgs" or a "Higgs Portal Scalar (HPS)". The dark higgs name alludes to the fact that S may be responsible for giving mass to a dark photon through spontaneous symmetry breaking in the dark sector.

- The neutrino portal:
$$\mathcal{L} = y \bar{L} \tilde{H} N, \tag{2.3}$$
where N is a new fermion that is a singlet under the Standard Model gauge group. This is precisely the same physics we discussed in the first part of the course, where $N = \nu_R$ is a right-handed neutrino.

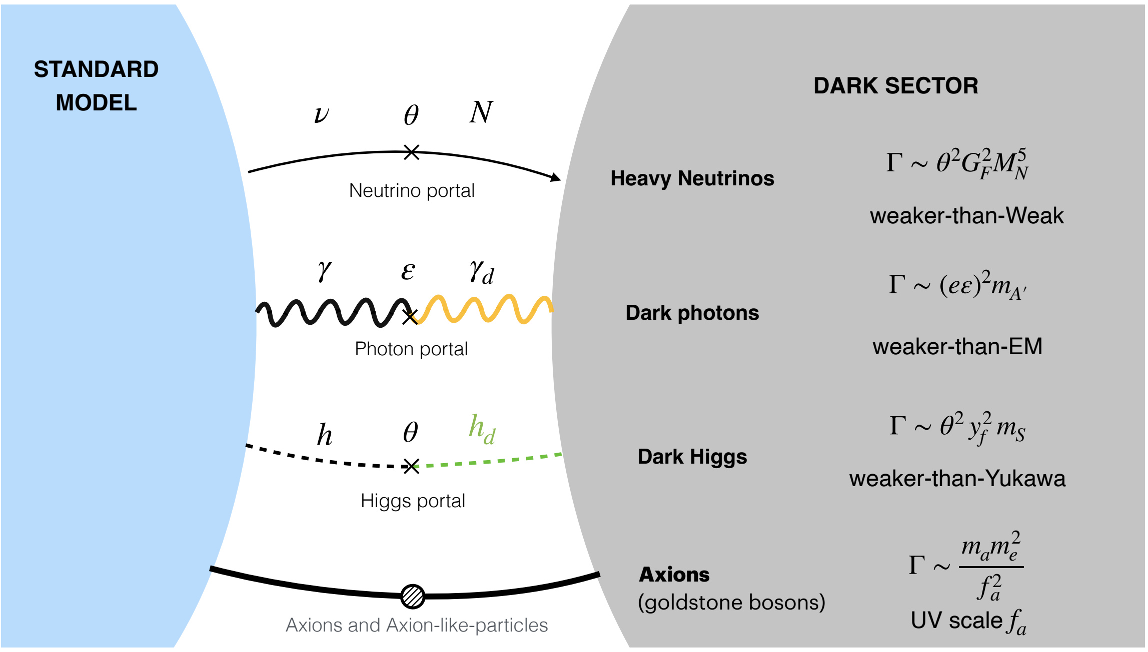

Figure 1: Examples of portals connecting the Standard Model to a dark sector along with the estimated decay rates of the new particles.

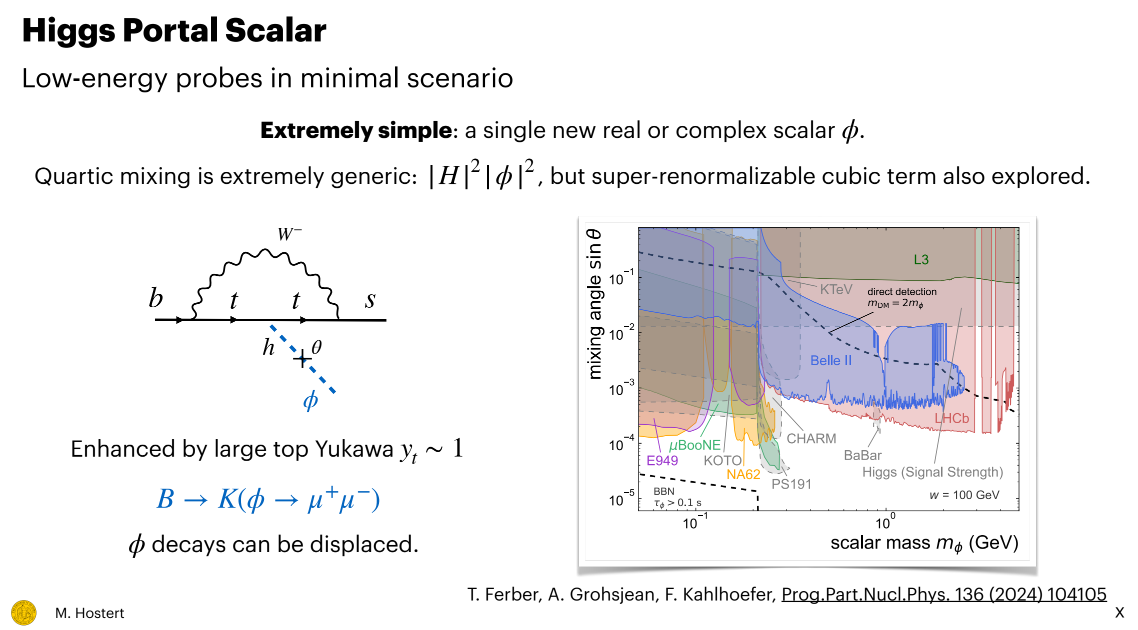

Figure 2: The Higgs portal scalar (HPS) can be produced in the decay of a meson, and then decay to a pair of leptons. The HPS can also be produced in the decay of the Higgs boson, and then decay to a pair of leptons.

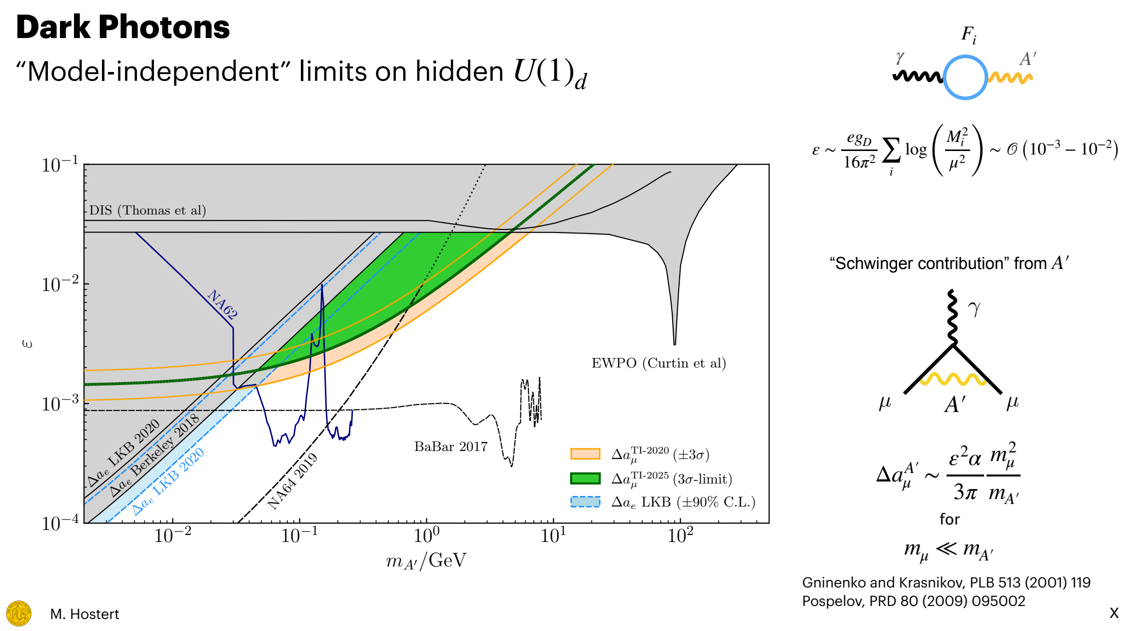

Figure 3: The dark photon parameter space.